COGNIZANT

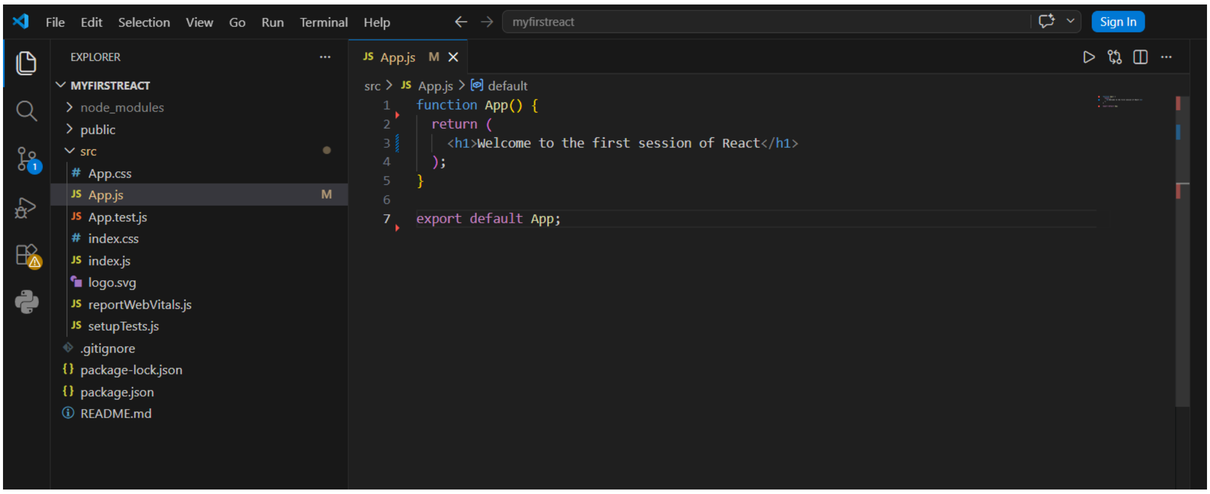
Mandatory Hands-on Exercises

-Khushi Ahi

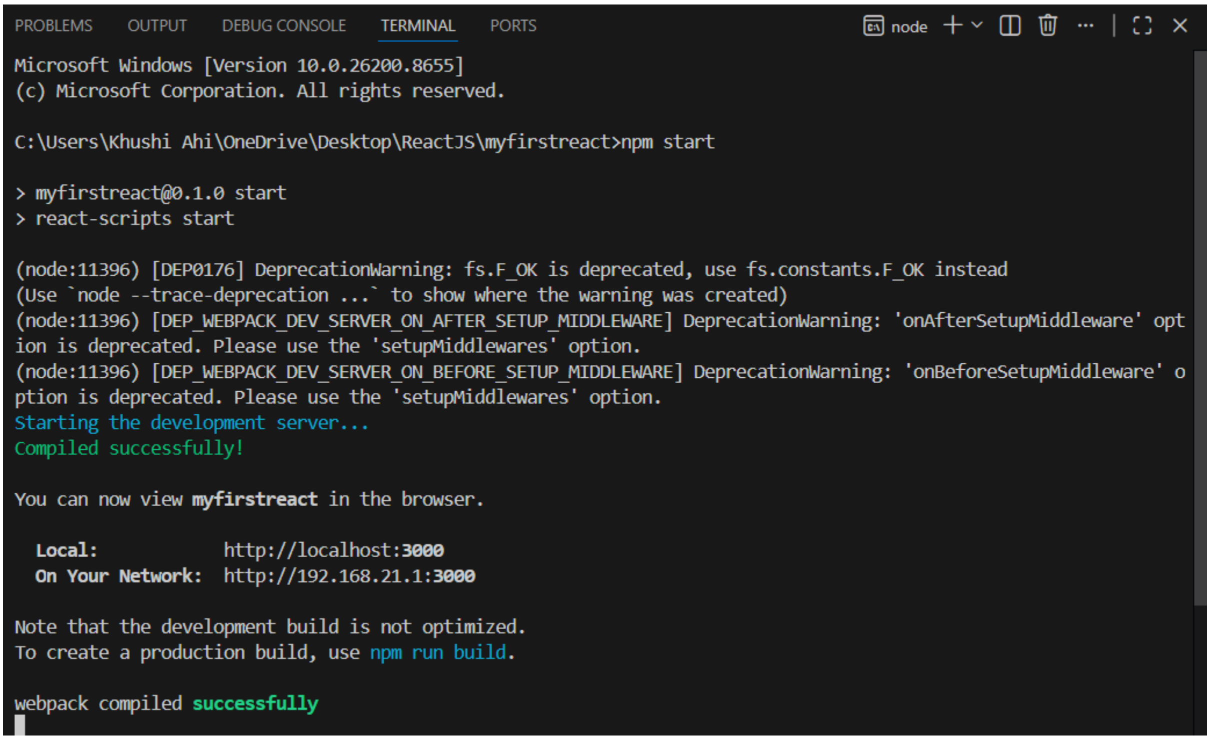
TOPIC: React

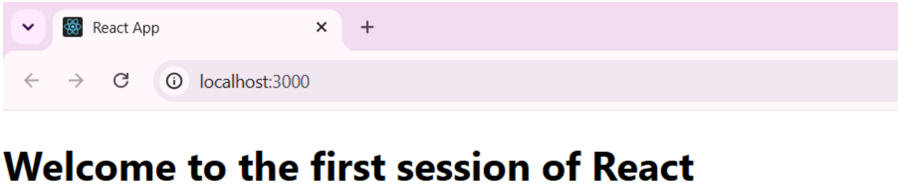
<u>Exercise 1: Creating the First React App</u>

CODE



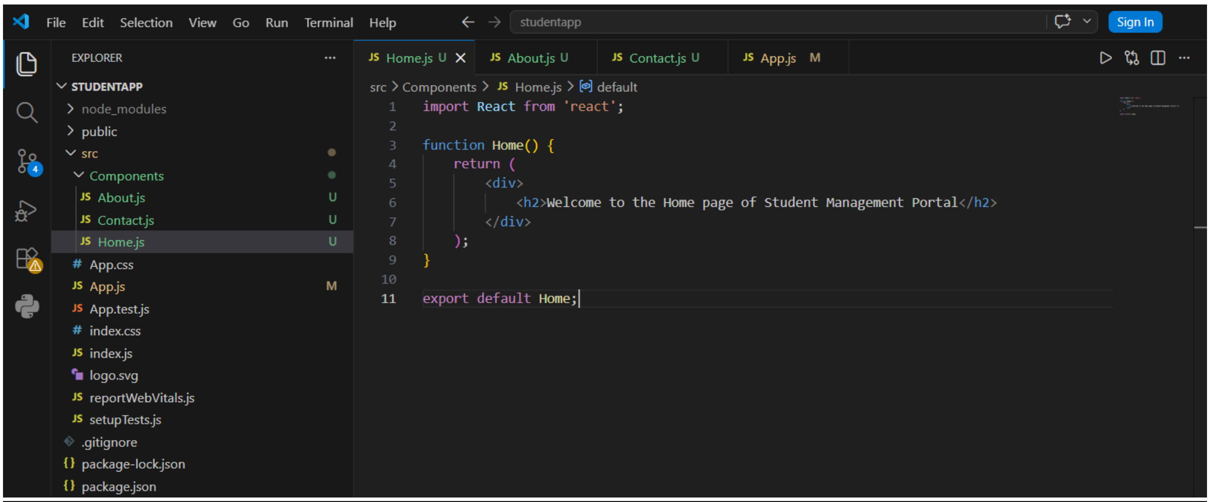
OUTPUT

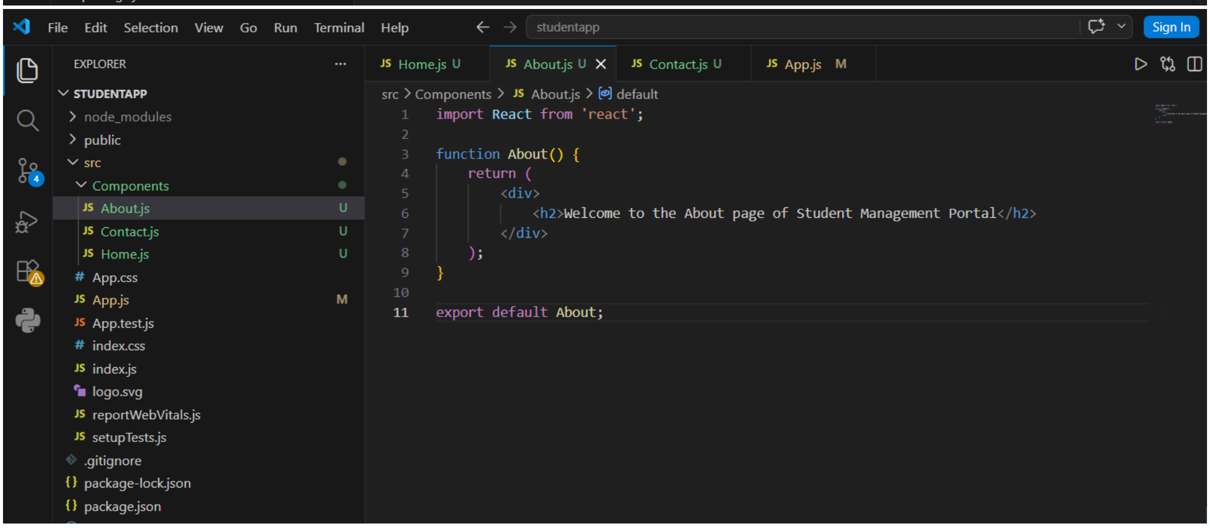


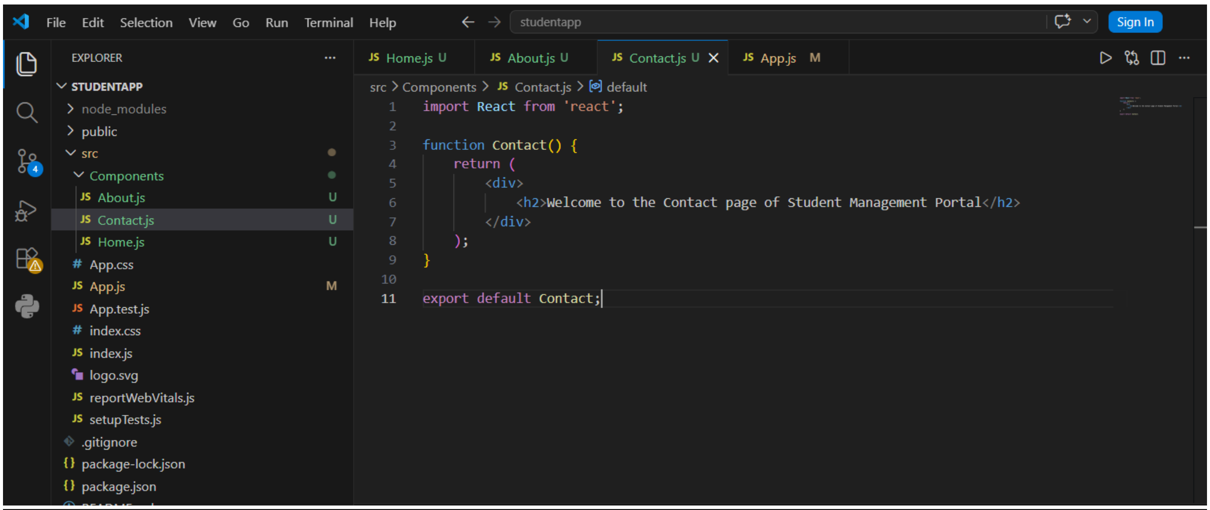


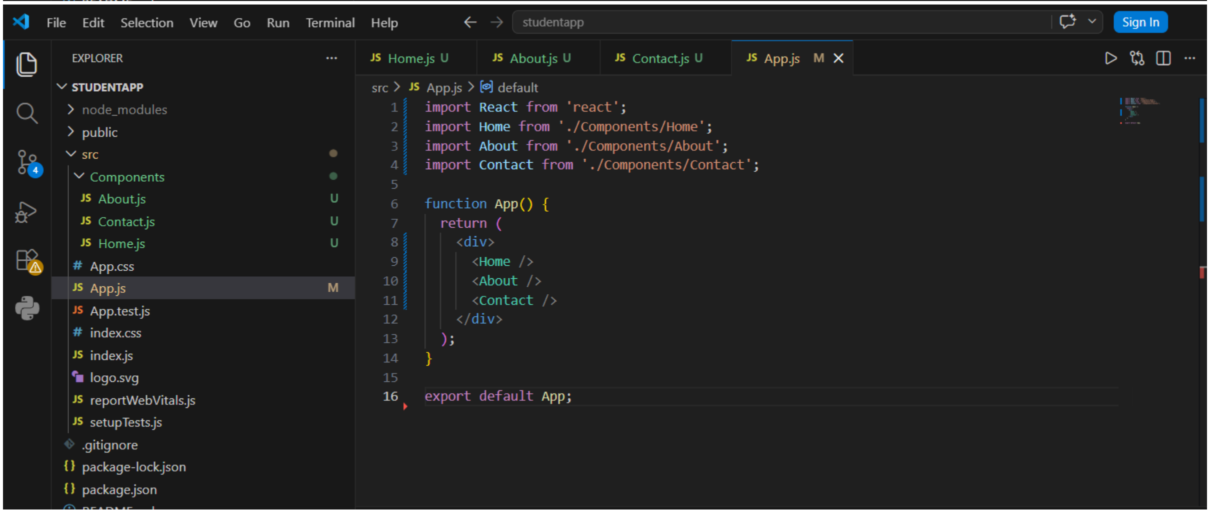
Exercise 2: Working with components

CODE






File Edit Selection View Go Run Terminal Help
studentapp
Sign In
EXPLORER
STUDENTAPP
node_modules
public
src
Components
About.js
Contact.js
Home.js
App.css
App.js
App.test.js
index.css
index.js
logo.svg
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json
Home.js U
About.js U
Contact.js U
App.js M
src > Components > Contact.js > default
import React from 'react';
function Contact() {
return (
<div>
<h2>Welcome to the Contact page of Student Management Portal</h2>
</div>
);
}
export default Contact;


File Edit Selection View Go Run Terminal Help
studentapp
Sign In
EXPLORER
STUDENTAPP
node_modules
public
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Components
About.js
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App.test.js
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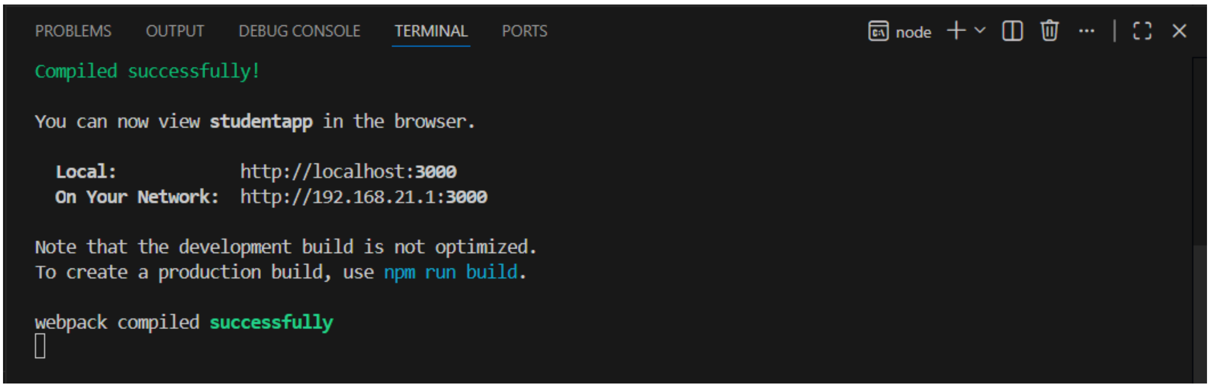
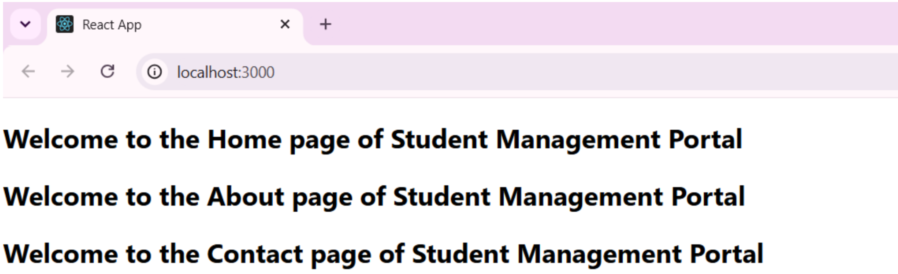
index.js
logo.svg
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json
Home.js U
About.js U
Contact.js U
App.js M
src > App.js > default
import React from 'react';
import Home from './Components/Home';
import About from './Components/About';
import Contact from './Components/Contact';
function App() {
return (
<div>
<Home />
<About />
<Contact />
</div>
);
}
export default App;

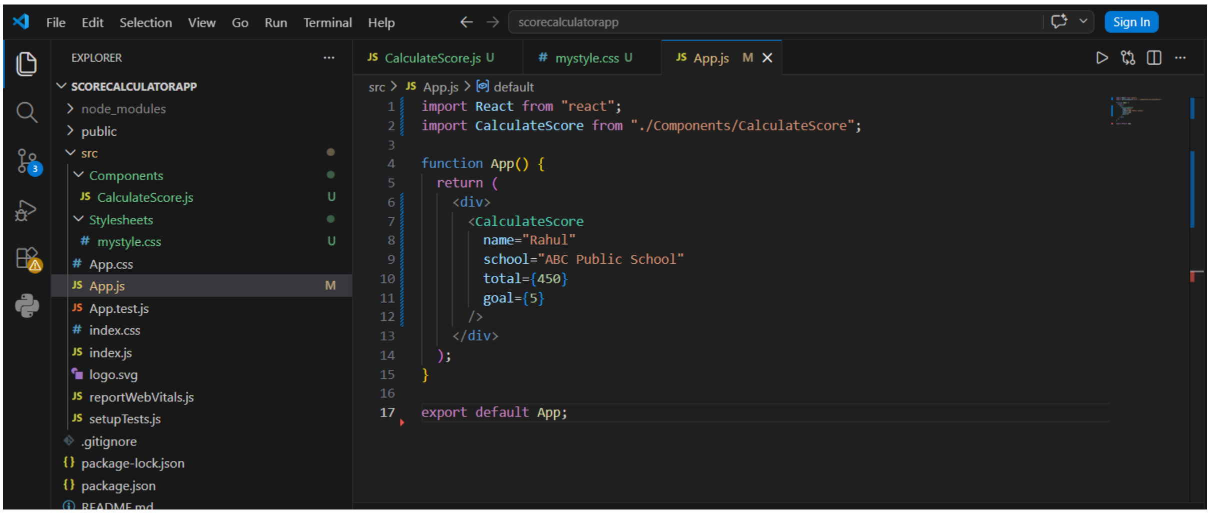
OUTPUT


PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
node
Compiled successfully!
You can now view studentapp in the browser.
Local: http://localhost:3000
On Your Network: http://192.168.21.1:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully

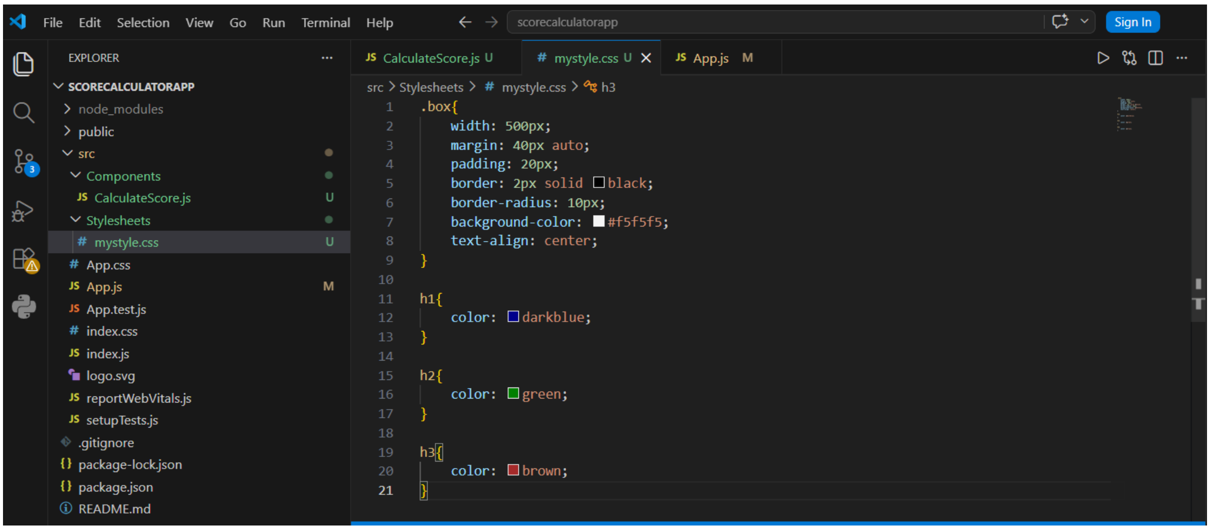


Exercise 3: Function Components and Styling

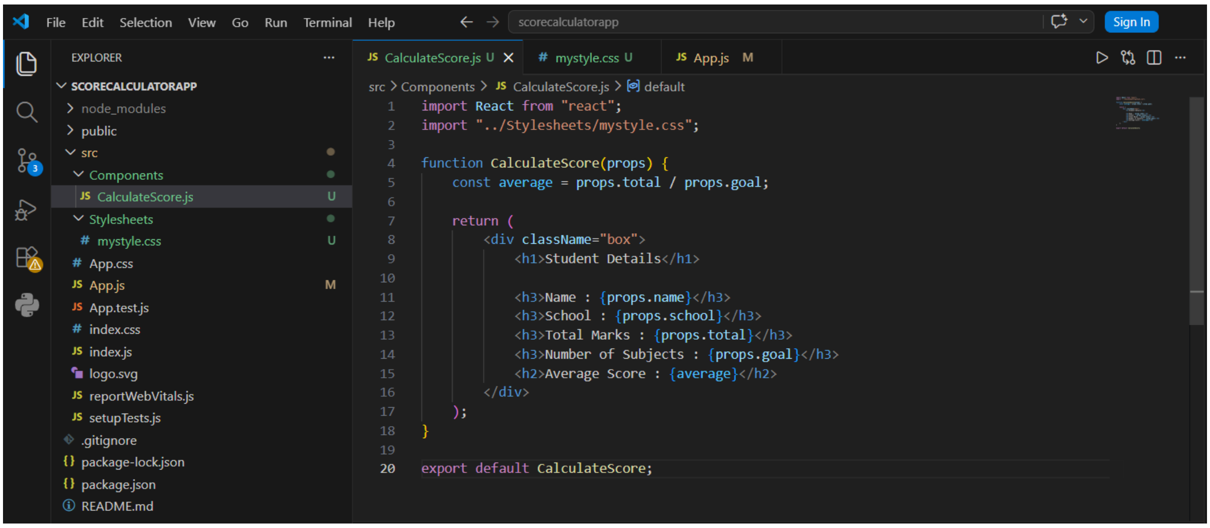
CODE



```
import React from "react";
import CalculateScore from "./Components/CalculateScore";

function App() {
  return (
    <div>
      <CalculateScore
        name="Rahul"
        school="ABC Public School"
        total={450}
        goal={5}
      />
    </div>
  );
}

export default App;
```

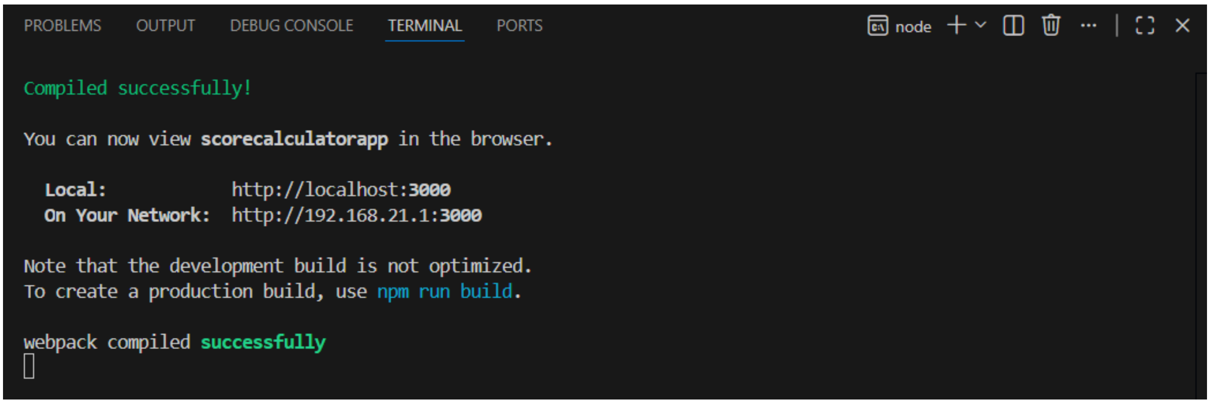


```
.box{
    width: 500px;
    margin: 40px auto;
    padding: 20px;
    border: 2px solid black;
    border-radius: 10px;
    background-color: #f5f5f5;
    text-align: center;
}

h1{
    color: darkblue;
}

h2{
    color: green;
}

h3{
    color: brown;
}
```

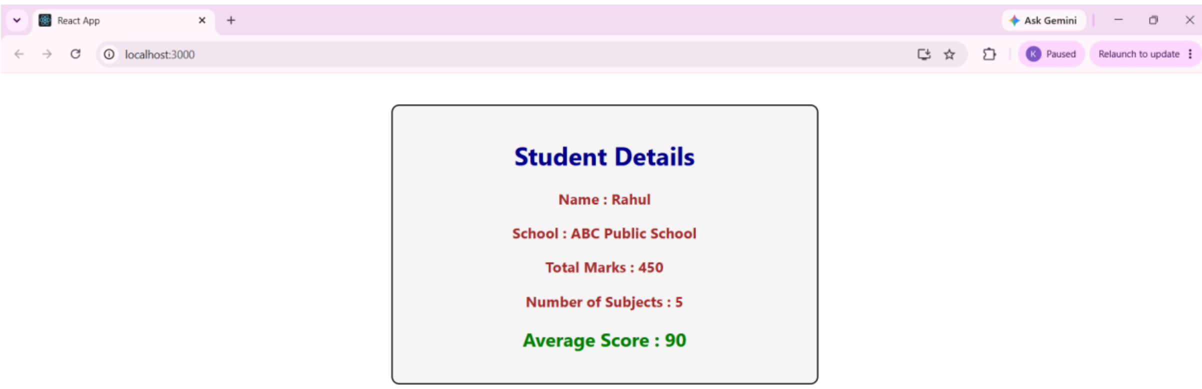


File Edit Selection View Go Run Terminal Help
scorecalculatorapp
Sign In
EXPLORER
SCORECALCULATORAPP
node_modules
public
src
Components
CalculateScore.js
Stylesheets
mystyle.css
App.css
App.js
App.test.js
index.css
index.js
logo.svg
reportWebVitals.js
setupTests.js
.gitignore
package-lock.json
package.json
README.md
CalculateScore.js U
mystyle.css U
App.js M
src > Components > CalculateScore.js > default
import React from "react";
import "../Stylesheets/mystyle.css";
function CalculateScore(props) {
const average = props.total / props.goal;
return (
<div className="box">
<h1>Student Details</h1>
<h3>Name : {props.name}</h3>
<h3>School : {props.school}</h3>
<h3>Total Marks : {props.total}</h3>
<h3>Number of Subjects : {props.goal}</h3>
<h2>Average Score : {average}</h2>
</div>
);
}
export default CalculateScore;

OUTPUT



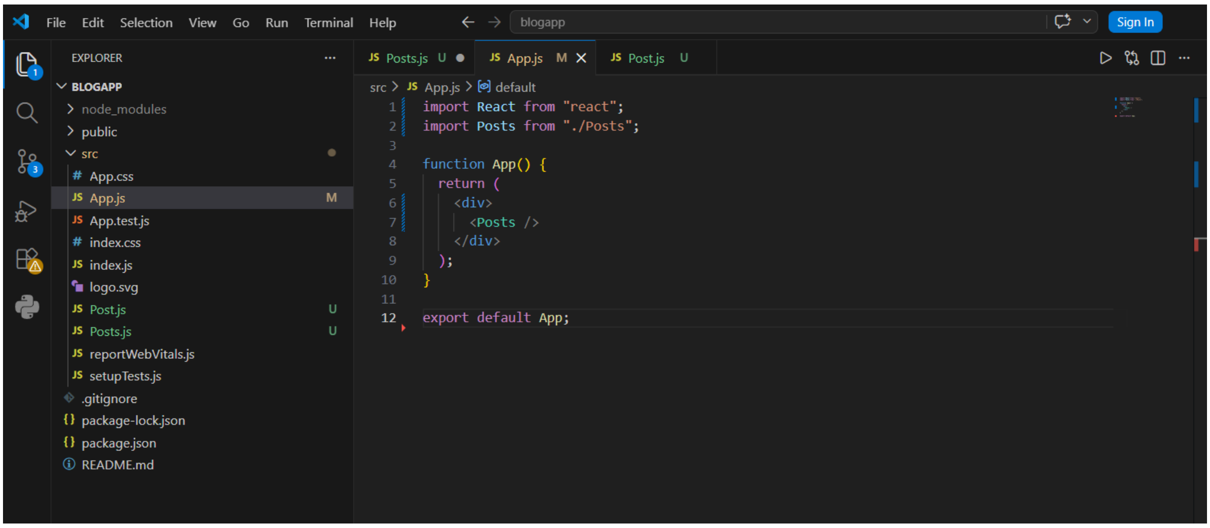
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
node
Compiled successfully!
You can now view scorecalculatorapp in the browser.
Local: http://localhost:3000
On Your Network: http://192.168.21.1:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully

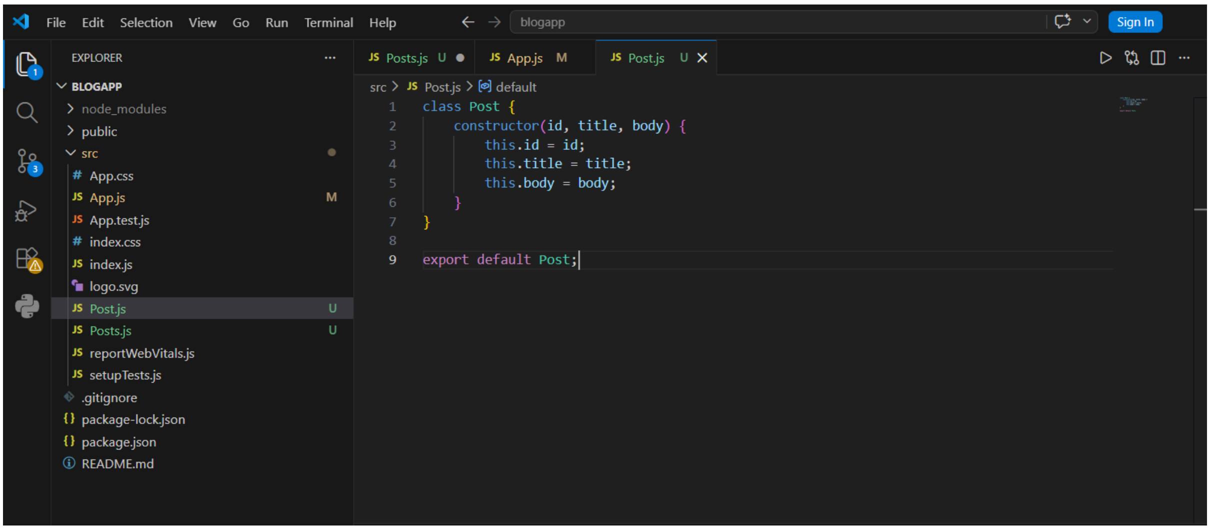


React App
Ask Gemini
localhost:3000
Paused
Relaunch to update
Student Details
Name : Rahul
School : ABC Public School
Total Marks : 450
Number of Subjects : 5
Average Score : 90

Exercise 4: Component Lifecycle Methods

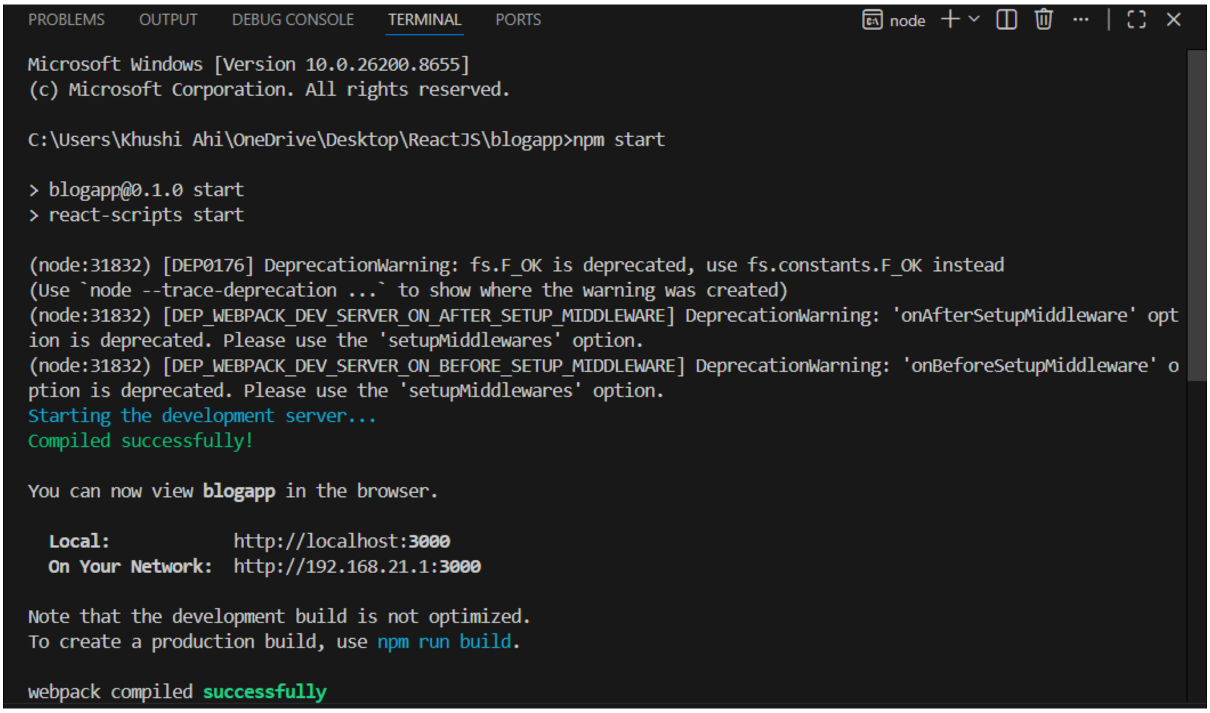
CODE

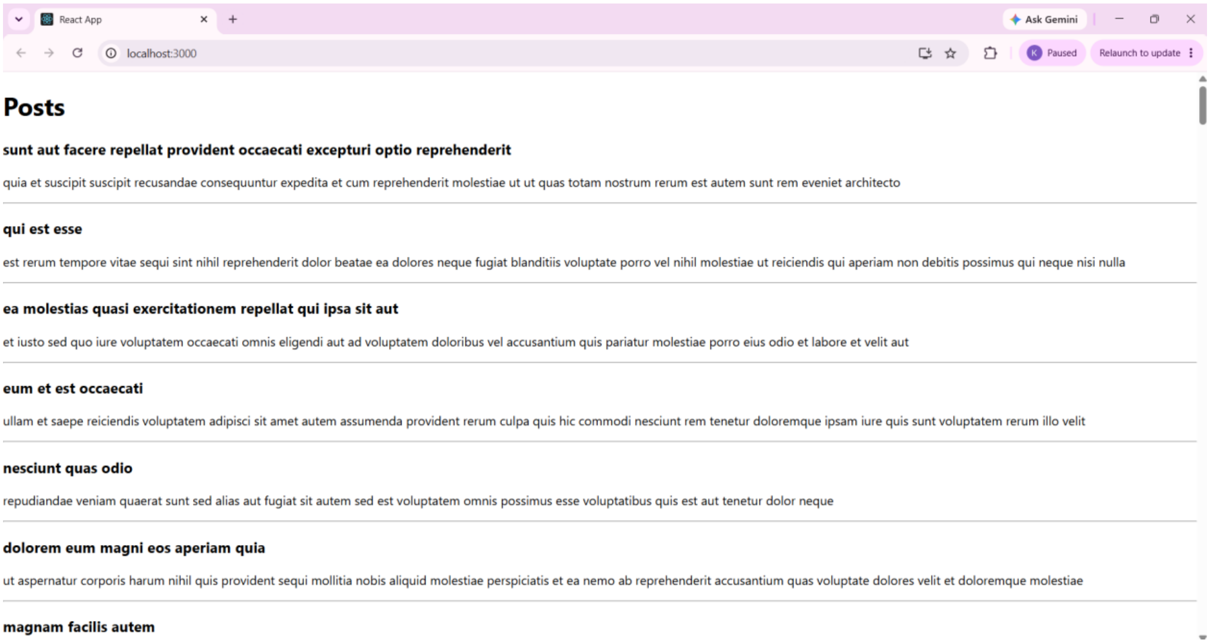




```
import React from "react";
import Post from "./Posts";

class Posts extends React.Component {

    constructor(props) {
        super(props);

        this.state = {
            posts: []
        };
    }

    loadPosts() {
        fetch("https://jsonplaceholder.typicode.com/posts")
            .then(response => response.json())
            .then(data => {
                const posts = data.map(
                    item => new Post(item.id, item.title, item.body)
                );

                this.setState({
                    posts: posts
                });
            });
    }

    componentDidMount() {
        this.loadPosts();
    }

    componentDidCatch(error, info) {
        alert("An error occurred: " + error);
    }

    render() {
        return (
            <div>
                <h1>Posts</h1>

                {
                    this.state.posts.map(post => (
                        <div key={post.id}>
                            <h3>{post.title}</h3>
                            <p>{post.body}</p>
                            <hr />
                        </div>
                    ))
                }

            </div>
        );
    }
}

export default Posts;
```

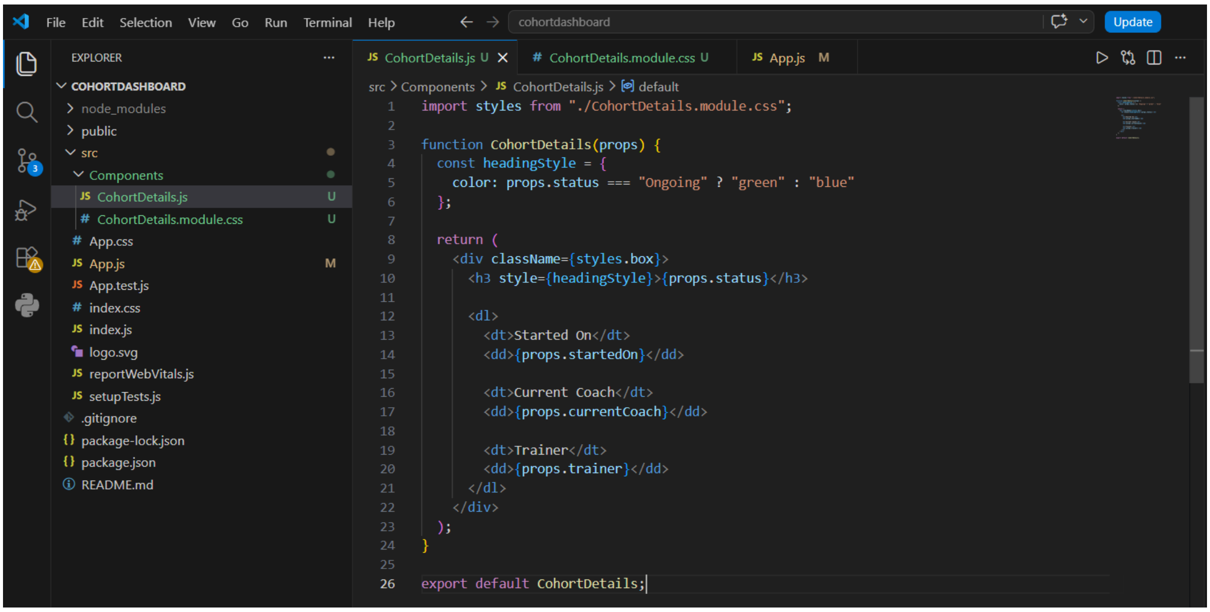
OUTPUT

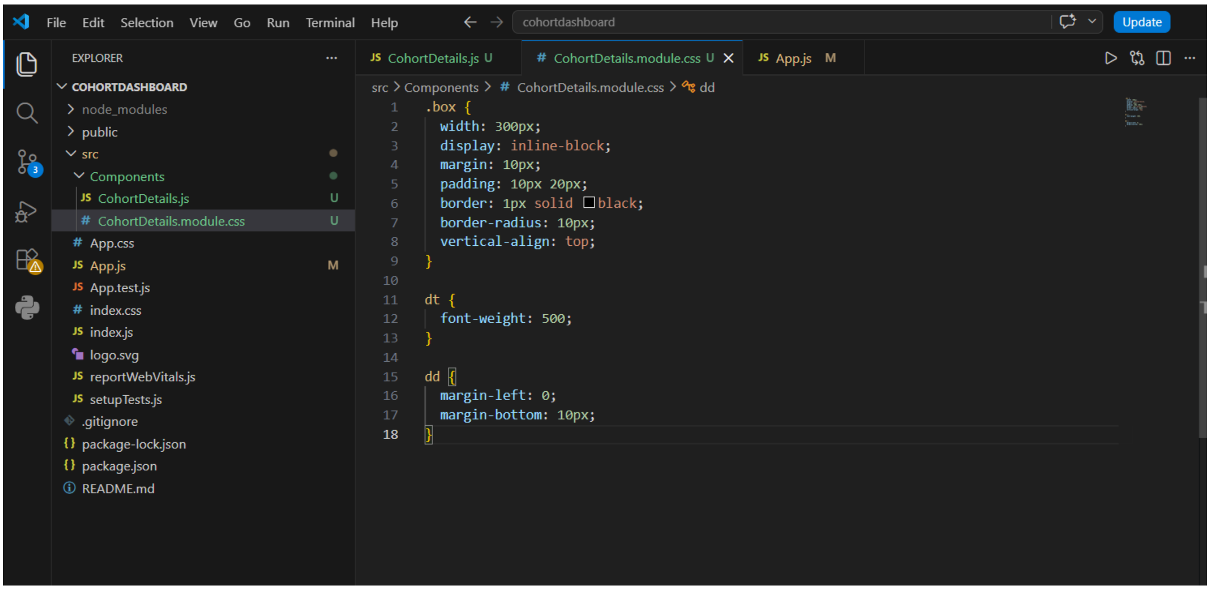

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
node
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Khushi Ahi\OneDrive\Desktop\ReactJS\blogapp>npm start
> blogapp@0.1.0 start
> react-scripts start
(node:31832) [DEP0176] DeprecationWarning: fs.F_OK is deprecated, use fs.constants.F_OK instead
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:31832) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:31832) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!
You can now view blogapp in the browser.
Local: http://localhost:3000
On Your Network: http://192.168.21.1:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully


React App
Ask Gemini
localhost:3000
Paused
Relaunch to update
Posts
sunt aut facere repellat provident occaecati excepturi optio reprehenderit
quia et suscipit suscipit recusandae consequuntur expedita et cum reprehenderit molestiae ut ut quas totam nostrum rerum est autem sunt rem eveniet architecto
qui est esse
est rerum tempore vitae sequi sint nihil reprehenderit dolor beatae ea dolores neque fugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis qui aperiam non debitis possimus qui neque nisi nulla
ea molestias quasi exercitationem repellat qui ipsa sit aut
et iusto sed quo iure voluptatem occaecati omnis eligendi aut ad voluptatem doloribus vel accusantium quis pariatur molestiae porro eius odio et labore et velit aut
eum et est occaecati
ullam et saepe reiciendis voluptatem adipisci sit amet autem assumenda provident rerum culpa quis hic commodi nesciunt rem tenetur doloremque ipsam iure quis sunt voluptatem rerum illo velit
nesciunt quas odio
repudiandae veniam quaerat sunt sed alias aut fugiat sit autem sed est voluptatem omnis possimus esse voluptatibus quis est aut tenetur dolor neque
dolorem eum magni eos aperiam quia
ut aspernatur corporis harum nihil quis provident sequi mollitia nobis aliquid molestiae perspiciatis et ea nemo ab reprehenderit accusantium quas voluptate dolores velit et doloremque molestiae
magnam facilis autem

Exercise 5: Styling React Components using CSS Modules

CODE

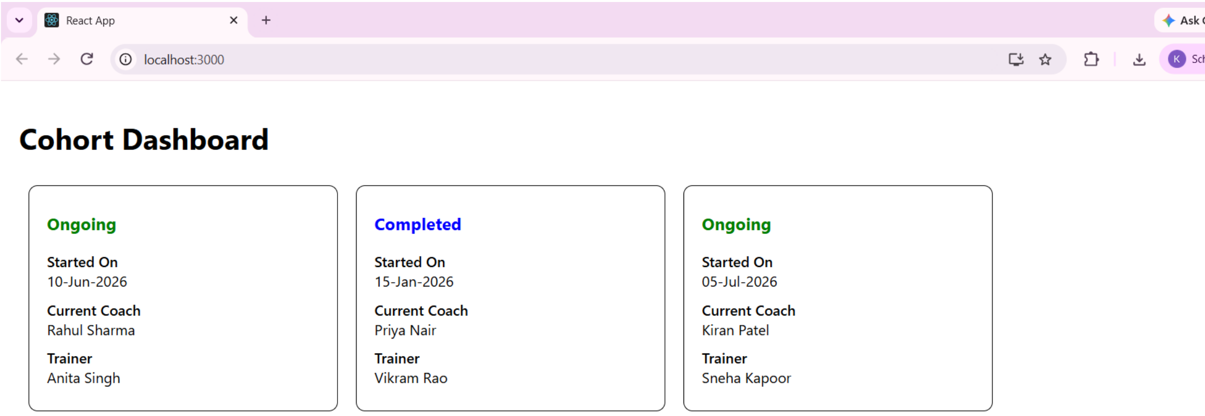




```jsx
import "./App.css";
import CohortDetails from "./Components/CohortDetails";

function App() {
  return (
    <div style={{ padding: "20px" }}>
      <h1>Cohort Dashboard</h1>

      <CohortDetails
        status="Ongoing"
        startedOn="10-Jun-2026"
        currentCoach="Rahul Sharma"
        trainer="Anita Singh"
      />

      <CohortDetails
        status="Completed"
        startedOn="15-Jan-2026"
        currentCoach="Priya Nair"
        trainer="Vikram Rao"
      />

      <CohortDetails
        status="Ongoing"
        startedOn="05-Jul-2026"
        currentCoach="Kiran Patel"
        trainer="Sneha Kapoor"
      />
    </div>
  );
}

export default App;
```

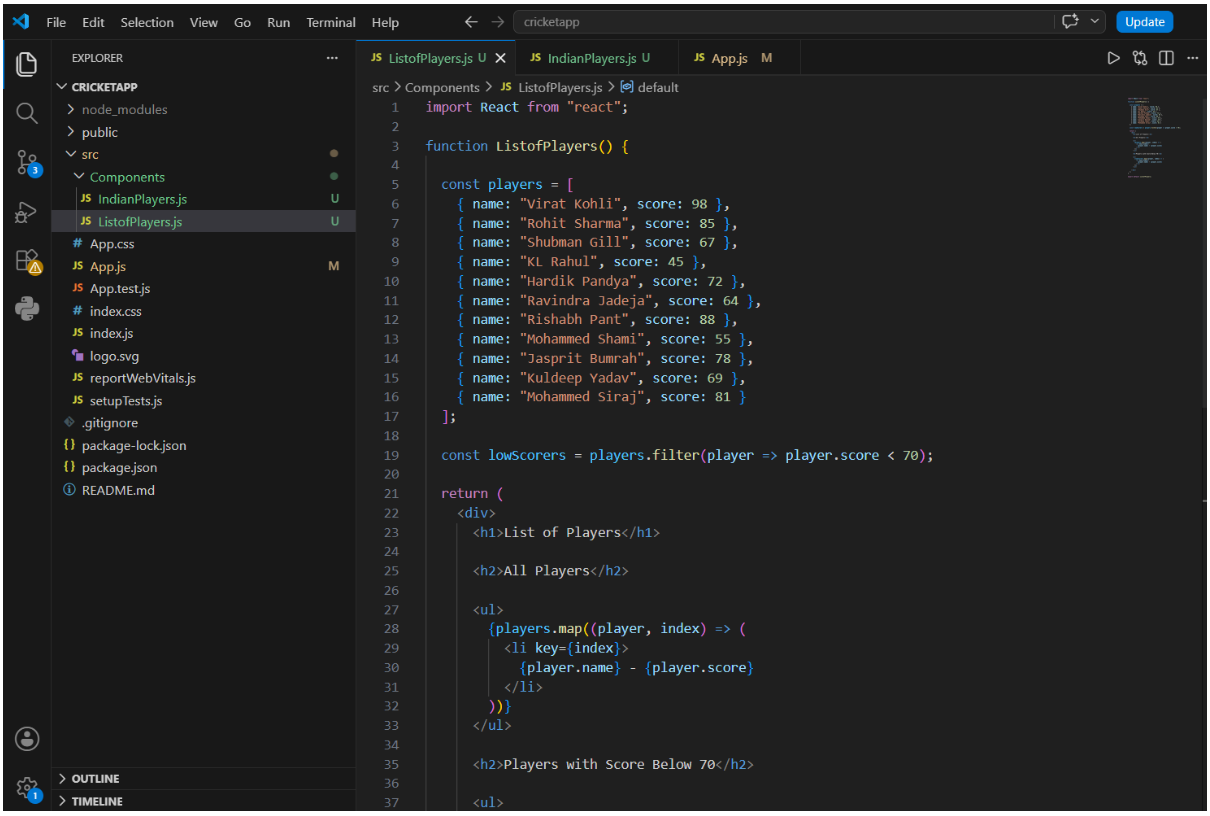
OUTPUT

```
(Use `node --trace-deprecation ...` to show where the warning was created)
(node:6896) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:6896) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view cohortdashboard in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.21.1:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```



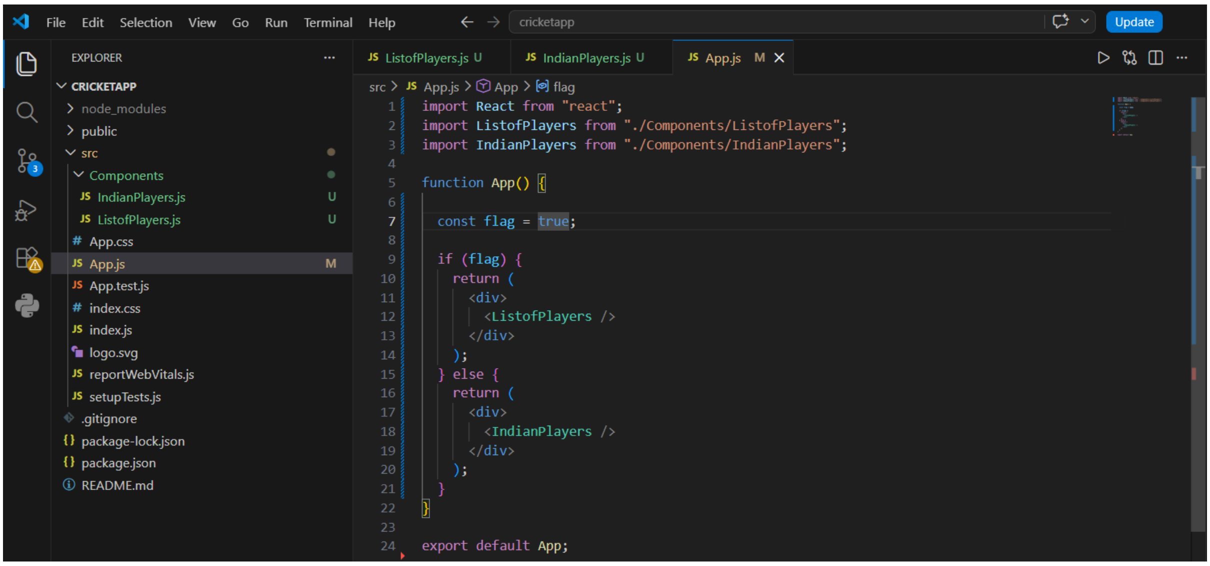
<u>Exercise 9: Cricket App</u>

CODE




```jsx
import React from "react";

function ListofPlayers() {

  const players = [
    { name: "Virat Kohli", score: 98 },
    { name: "Rohit Sharma", score: 85 },
    { name: "Shubman Gill", score: 67 },
    { name: "KL Rahul", score: 45 },
    { name: "Hardik Pandya", score: 72 },
    { name: "Ravindra Jadeja", score: 64 },
    { name: "Rishabh Pant", score: 88 },
    { name: "Mohammed Shami", score: 55 },
    { name: "Jasprit Bumrah", score: 78 },
    { name: "Kuldeep Yadav", score: 69 },
    { name: "Mohammed Siraj", score: 81 }
  ];

  const lowScorers = players.filter(player => player.score < 70);

  return (
    <div>
      <h1>List of Players</h1>

      <h2>All Players</h2>

      <ul>
        {players.map((player, index) => (
          <li key={index}>
            {player.name} - {player.score}
          </li>
        ))}
      </ul>

      <h2>Players with Score Below 70</h2>

      <ul>
        {lowScorers.map((player, index) => (
          <li key={index}>
            {player.name} - {player.score}
          </li>
        ))}
      </ul>

    </div>
  );
}

export default ListofPlayers;
```

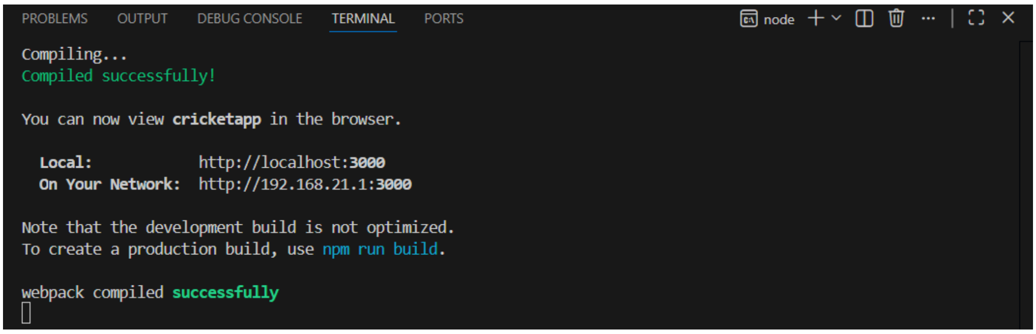
```
import React from "react";

function IndianPlayers() {

  // Destructuring
  const players = [
    "Virat Kohli",
    "Rohit Sharma",
    "Shubman Gill",
    "KL Rahul",
    "Hardik Pandya",
    "Ravindra Jadeja"
  ];

  const [odd1, even1, odd2, even2, odd3, even3] = players;

  const oddTeam = [odd1, odd2, odd3];
  const evenTeam = [even1, even2, even3];

  // Merge using ES6 Spread Operator
  const T20players = [
    "Virat Kohli",
    "Rohit Sharma",
    "Hardik Pandya"
  ];

  const RanjiPlayers = [
    "Mayank Agarwal",
    "Cheteshwar Pujara",
    "Hanuma Vihari"
  ];

  const IndianPlayers = [...T20players, ...RanjiPlayers];

  return (
    <div>
      <h1>Indian Players</h1>

      <h2>Odd Team Players</h2>
      <ul>
        {oddTeam.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>

      <h2>Even Team Players</h2>
      <ul>
        {evenTeam.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>

      <h2>Merged Players</h2>
      <ul>
        {IndianPlayers.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>
    </div>
  );
}

export default IndianPlayers;
```

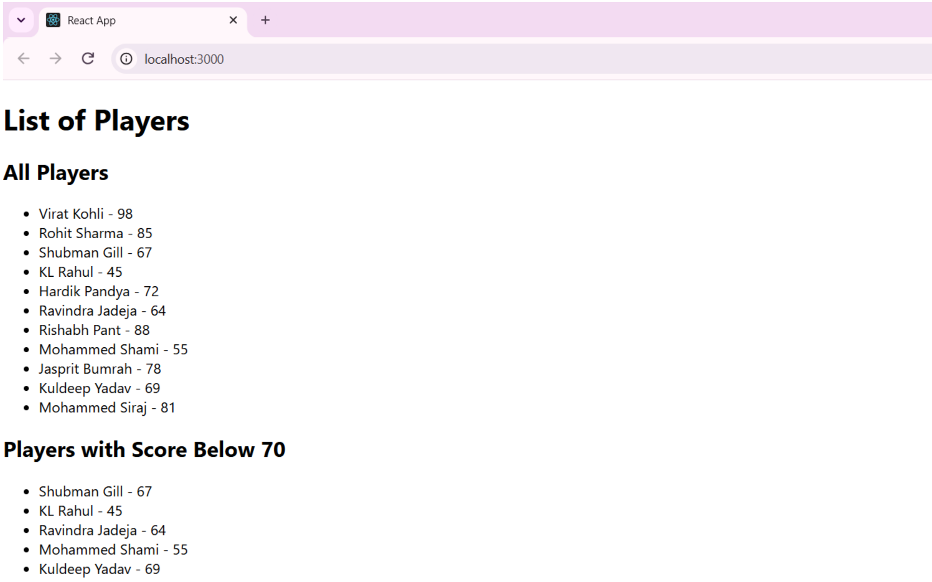


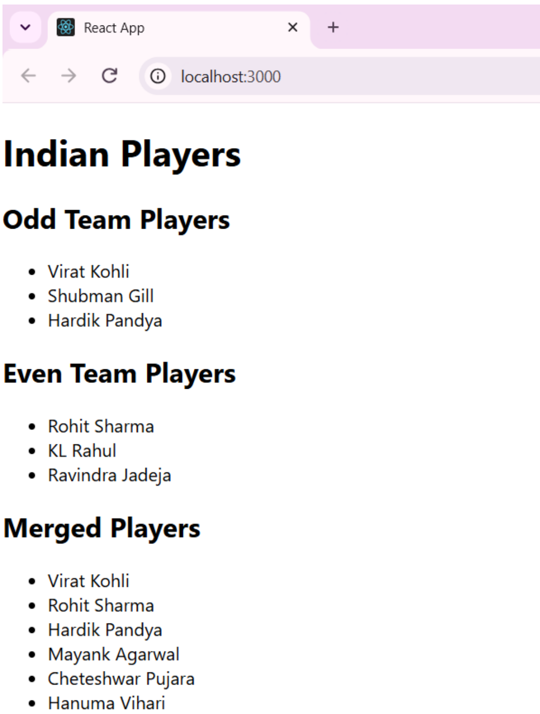
src > App.js > App > flag

```
import React from "react";
import ListofPlayers from "./Components/ListofPlayers";
import IndianPlayers from "./Components/IndianPlayers";

function App() {

  const flag = false;

  if (flag) {
    return (
      <div>
        <ListofPlayers />
      </div>
    );
  } else {
    return (
      <div>
        <IndianPlayers />
      </div>
    );
  }
}

export default App;
```

OUTPUT


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
node
Compiling...
Compiled successfully!
You can now view cricketapp in the browser.
Local: http://localhost:3000
On Your Network: http://192.168.21.1:3000
Note that the development build is not optimized.
To create a production build, use npm run build.
webpack compiled successfully


React App
localhost:3000
List of Players
All Players
Virat Kohli - 98
Rohit Sharma - 85
Shubman Gill - 67
KL Rahul - 45
Hardik Pandya - 72
Ravindra Jadeja - 64
Rishabh Pant - 88
Mohammed Shami - 55
Jasprit Bumrah - 78
Kuldeep Yadav - 69
Mohammed Siraj - 81
Players with Score Below 70
Shubman Gill - 67
KL Rahul - 45
Ravindra Jadeja - 64
Mohammed Shami - 55
Kuldeep Yadav - 69


React App
localhost:3000
Indian Players
Odd Team Players
Virat Kohli
Shubman Gill
Hardik Pandya
Even Team Players
Rohit Sharma
KL Rahul
Ravindra Jadeja
Merged Players
Virat Kohli
Rohit Sharma
Hardik Pandya
Mayank Agarwal
Cheteshwar Pujara
Hanuma Vihari

Exercise 10: Office Space Rental App

CODE

```javascript
import React from "react";

function App() {

  const officeSpaces = [
    {
      Name: "DBS",
      Rent: 50000,
      Address: "Chennai",
      Image:
        "https://images.unsplash.com/photo-1497366754035-f200968a6e72?w=500"
    },
    {
      Name: "Regus",
      Rent: 75000,
      Address: "Bangalore",
      Image:
        "https://images.unsplash.com/photo-1497366412874-3415097a27e7?w=500"
    },
    {
      Name: "WeWork",
      Rent: 55000,
      Address: "Hyderabad",
      Image:
        "https://images.unsplash.com/photo-1497366811353-6870744d04b2?w=500"
    }
  ];

  return (
    <div style={{ margin: "30px" }}>

      <h1>Office Space, at Affordable Range</h1>

      {officeSpaces.map((office, index) => (

        <div key={index} style={{ marginBottom: "40px" }}>

          <img
            src={office.Image}
            alt="Office Space"
            width="300"
            height="200"
          />

          <h2>Name: {office.Name}</h2>

          <h3
            style={{
              color: office.Rent <= 60000 ? "red" : "green"
            }}
          >
            Rent: Rs. {office.Rent}
          </h3>

          <h3>Address: {office.Address}</h3>

        </div>

      ))}

    </div>
  );
}

export default App;
```

OUTPUT

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS node

```
Compiling...
Compiled successfully!

You can now view officespacerentalapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.21.1:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

React App

localhost:3000

Office Space, at Affordable Range

Name: DBS

Rent: Rs. 50000

Address: Chennai

React App

localhost:3000

Name: Regus

Rent: Rs. 75000

Address: Bangalore

Name: WeWork

Rent: Rs. 55000

Address: Hyderabad

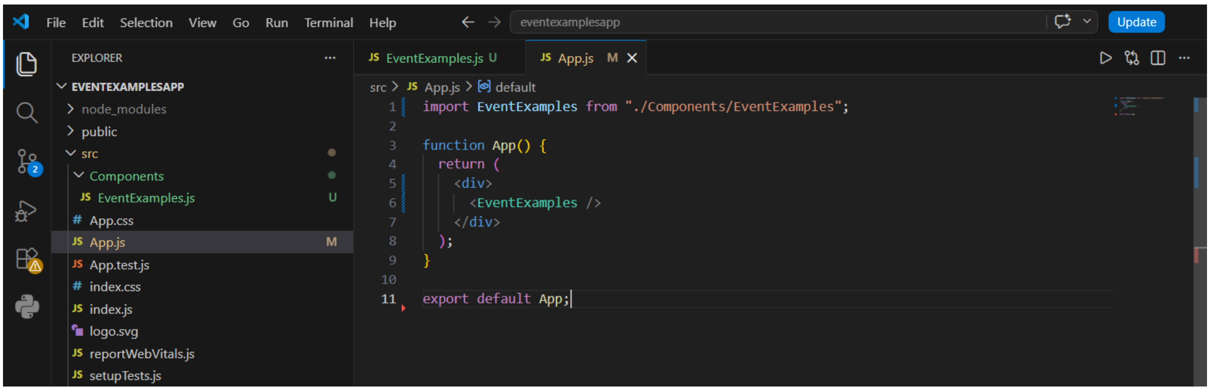
Exercise 11: React Event Handling

CODE

```jsx
import React, { useState } from "react";

function EventExamples() {
  const [count, setCount] = useState(5);

  const increment = () => {
    setCount(count + 1);
  };

  const decrement = () => {
    setCount(count - 1);
  };

  const sayHello = () => {
    alert("Hello! This is a static message.");
  };

  const handleIncrement = () => {
    increment();
    sayHello();
  };

  const sayWelcome = (msg) => {
    alert(msg);
  };

  const handlePress = () => {
    alert("I was clicked");
  };

  const [amount, setAmount] = useState("");
  const [currency, setCurrency] = useState("");

  const handleSubmit = (e) => {
    e.preventDefault();

    if (currency.toLowerCase() === "euro") {
```

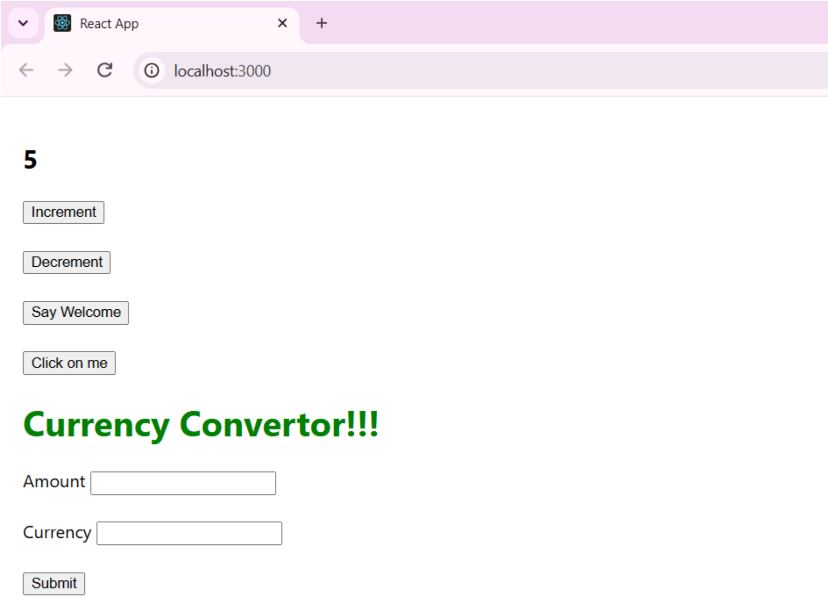
```jsx
      const result = (parseFloat(amount) / 90).toFixed(2);
      alert("Converting to Euro: " + result + " EUR");
    } else {
      alert("Enter Currency as Euro");
    }
  };

  return (
    <div style={{ padding: "20px" }}>
      <h2>{count}</h2>

      <button onClick={handleIncrement}>Increment</button>
      <br /><br />

      <button onClick={decrement}>Decrement</button>
      <br /><br />

      <button onClick={() => sayWelcome("Welcome")}>
        Say Welcome
      </button>
      <br /><br />

      <button onClick={handlePress}>Click on me</button>

      <h1 style={{ color: "green" }}>Currency Convertor!!!</h1>

      <form onSubmit={handleSubmit}>
        <div>
          <label>Amount </label>
          <input
            type="number"
            value={amount}
            onChange={(e) => setAmount(e.target.value)}
          />
        </div>
```

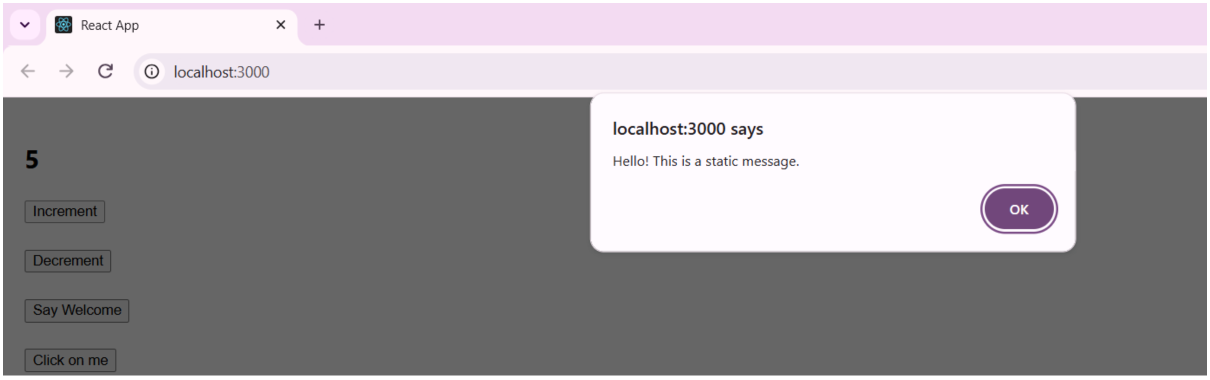
```
        <br />

        <div>
          <label>Currency </label>
          <input
            type="text"
            value={currency}
            onChange={(e) => setCurrency(e.target.value)}
          />
        </div>

        <br />

        <button type="submit">Submit</button>
      </form>
    </div>
  );
}

export default EventExamples;
```



OUTPUT

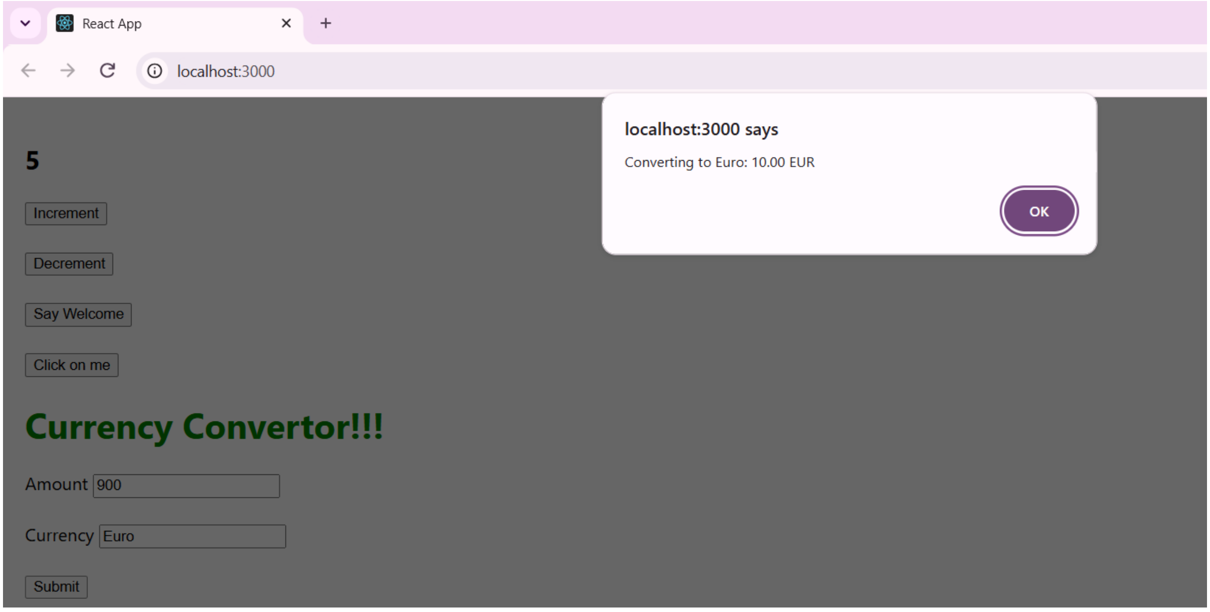
```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS                node

Compiling...
Compiled successfully!

You can now view eventexamplesapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.21.1:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

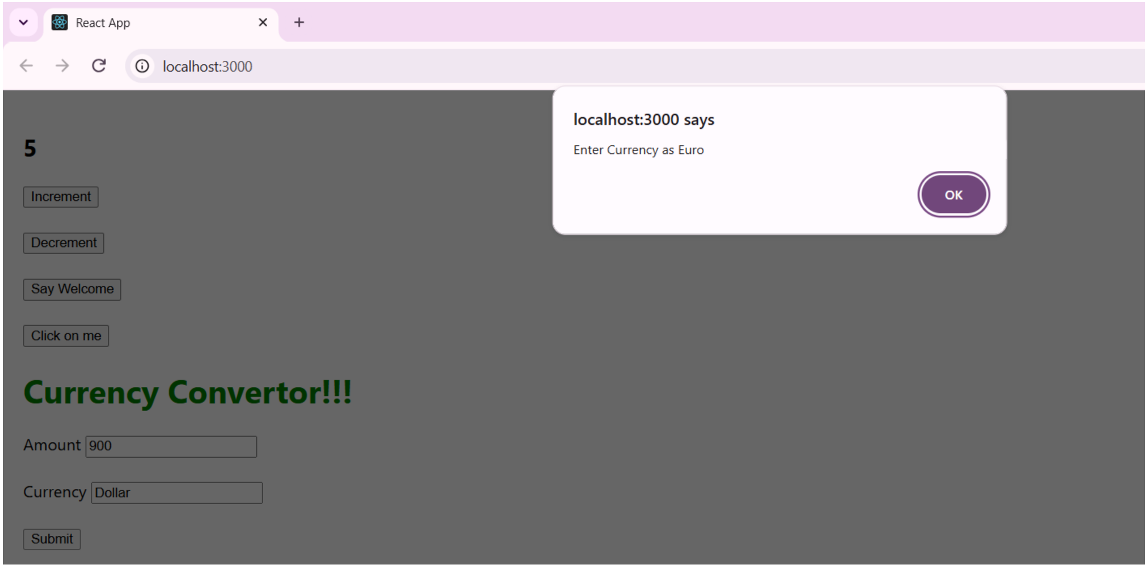

React App
localhost:3000
5
Increment
Decrement
Say Welcome
Click on me
Currency Convertor!!!
Amount
Currency
Submit


React App
localhost:3000
localhost:3000 says
Hello! This is a static message.
OK
5
Increment
Decrement
Say Welcome
Click on me


React App
localhost:3000
localhost:3000 says
Welcome
OK
5
Increment
Decrement
Say Welcome
Click on me

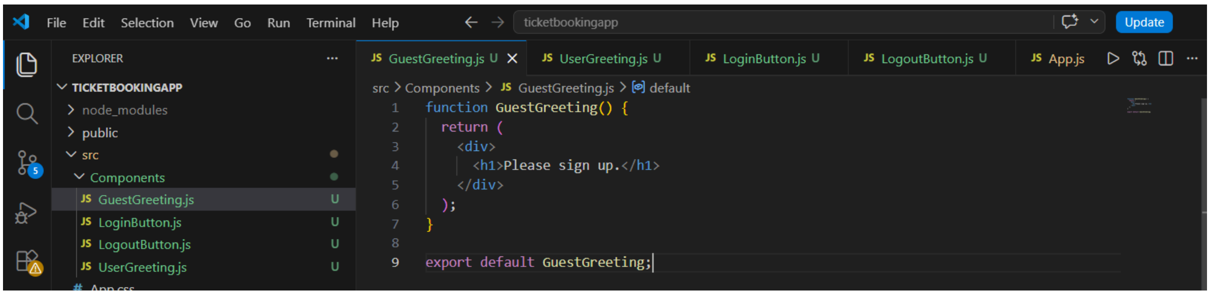

React App
localhost:3000
localhost:3000 says
I was clicked
OK
5
Increment
Decrement
Say Welcome
Click on me

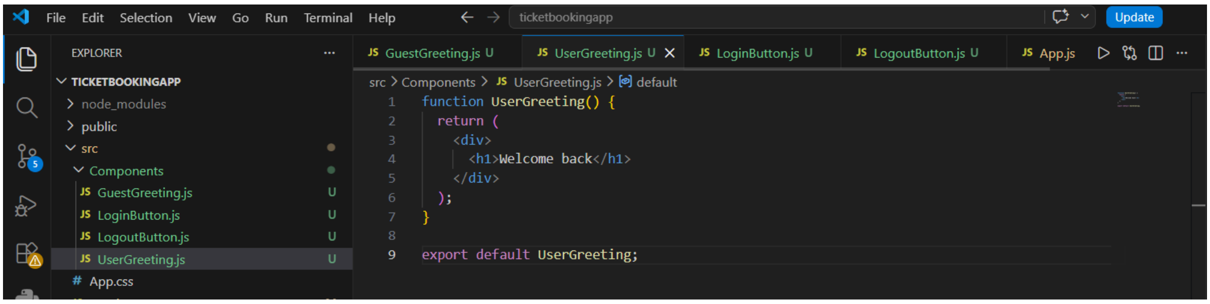




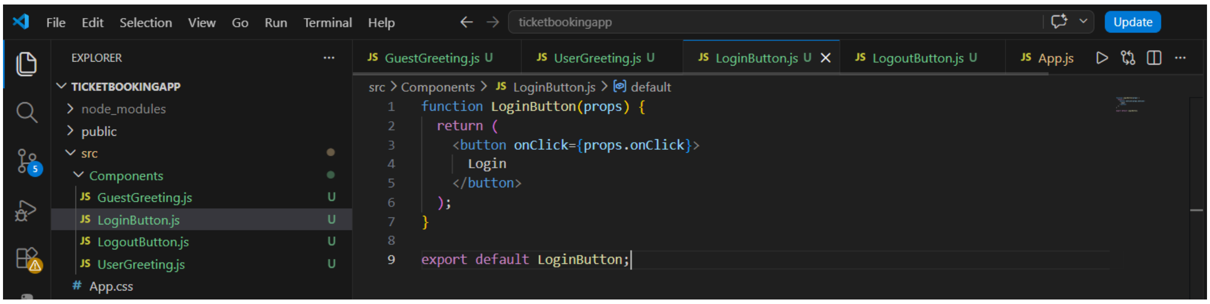
Exercise 12: Conditional Rendering in React

CODE

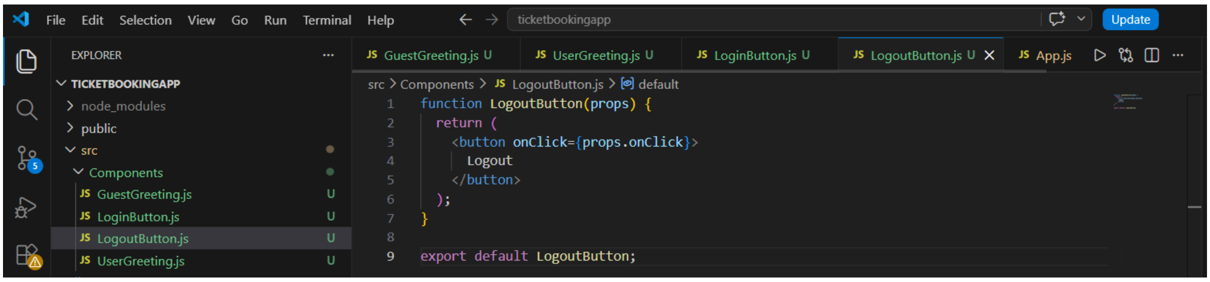




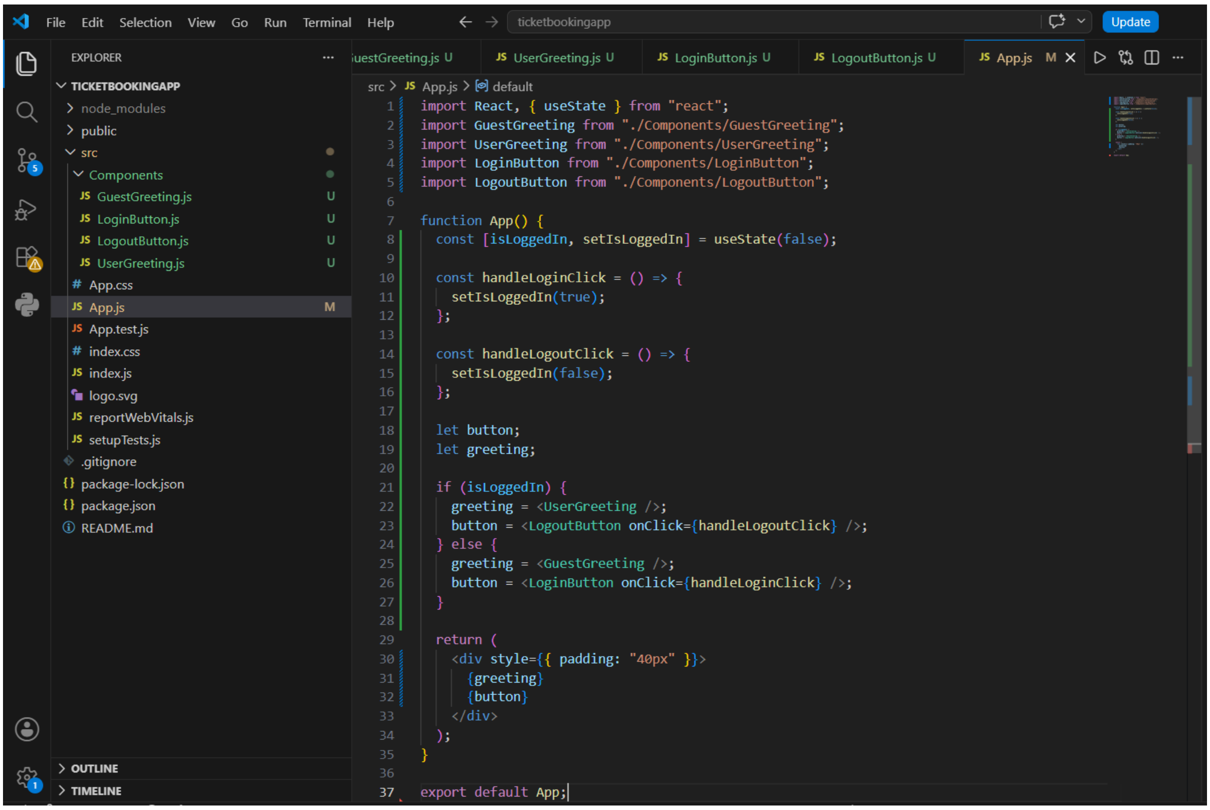
ticketbookingapp
src > Components > UserGreeting.js > default
function UserGreeting() {
return (
<div>
<h1>Welcome back</h1>
</div>
);
}
export default UserGreeting;



ticketbookingapp
src > Components > LoginButton.js > default
function LoginButton(props) {
return (
<button onClick={props.onClick}>
Login
</button>
);
}
export default LoginButton;

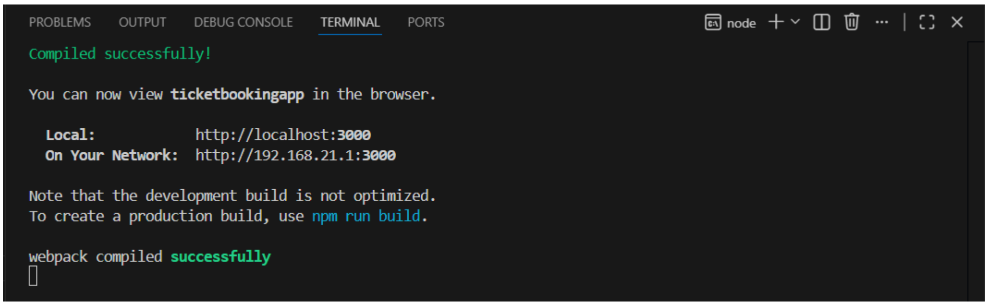


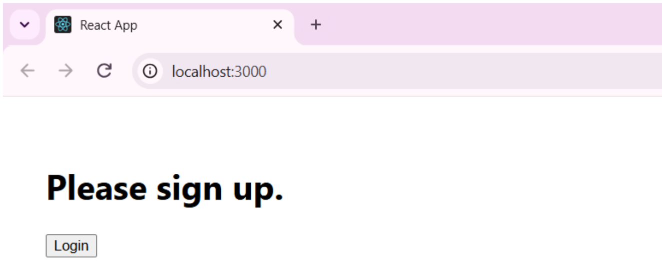
ticketbookingapp
src > Components > LogoutButton.js > default
function LogoutButton(props) {
return (
<button onClick={props.onClick}>
Logout
</button>
);
}
export default LogoutButton;

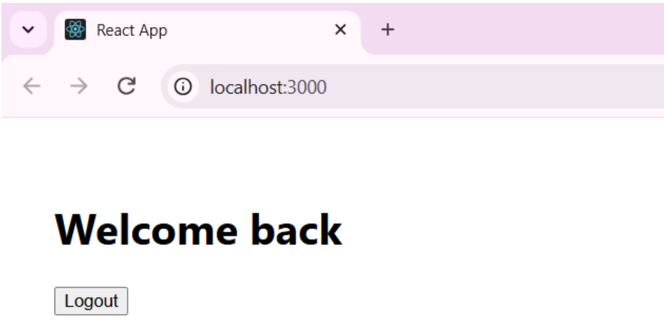


ticketbookingapp
src > App.js > default
import React, { useState } from "react";
import GuestGreeting from "./Components/GuestGreeting";
import UserGreeting from "./Components/UserGreeting";
import LoginButton from "./Components/LoginButton";
import LogoutButton from "./Components/LogoutButton";
function App() {
const [isLoggedIn, setIsLoggedIn] = useState(false);
const handleLoginClick = () => {
setIsLoggedIn(true);
};
const handleLogoutClick = () => {
setIsLoggedIn(false);
};
let button;
let greeting;
if (isLoggedIn) {
greeting = <UserGreeting />;
button = <LogoutButton onClick={handleLogoutClick} />;
} else {
greeting = <GuestGreeting />;
button = <LoginButton onClick={handleLoginClick} />;
}
return (
<div style={{ padding: "40px" }}>
{greeting}
{button}
</div>
);
}
export default App;

OUTPUT

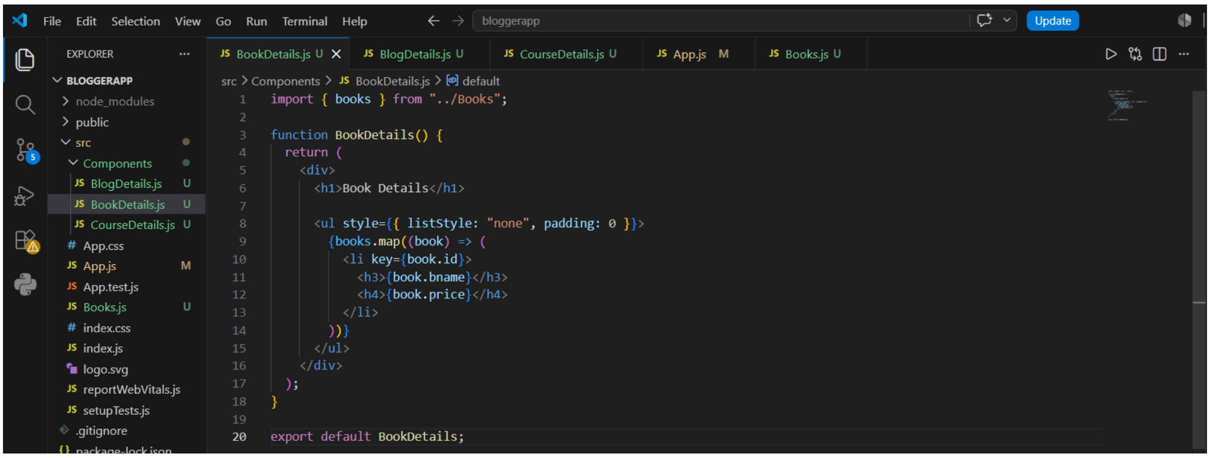


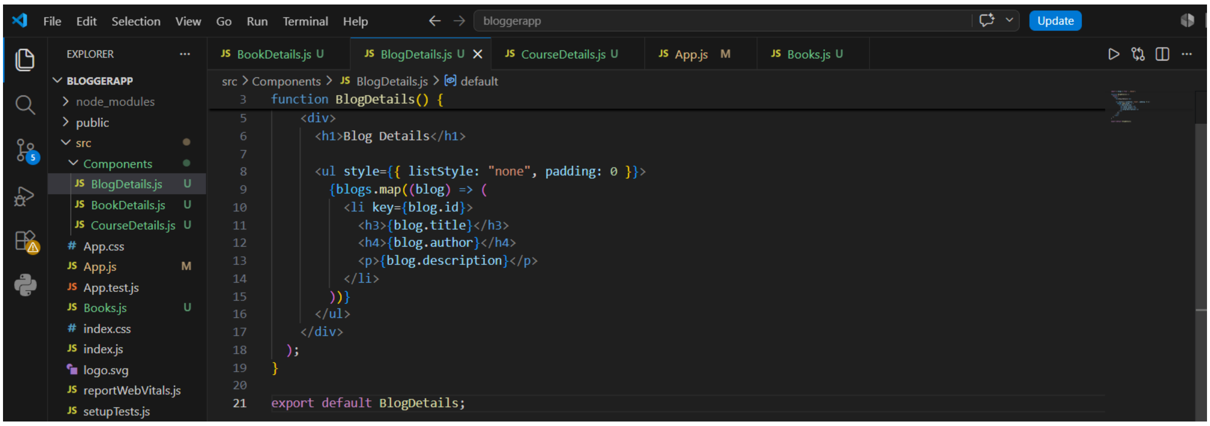




Exercise 13: Lists, Keys, map() and Conditional Rendering

CODE

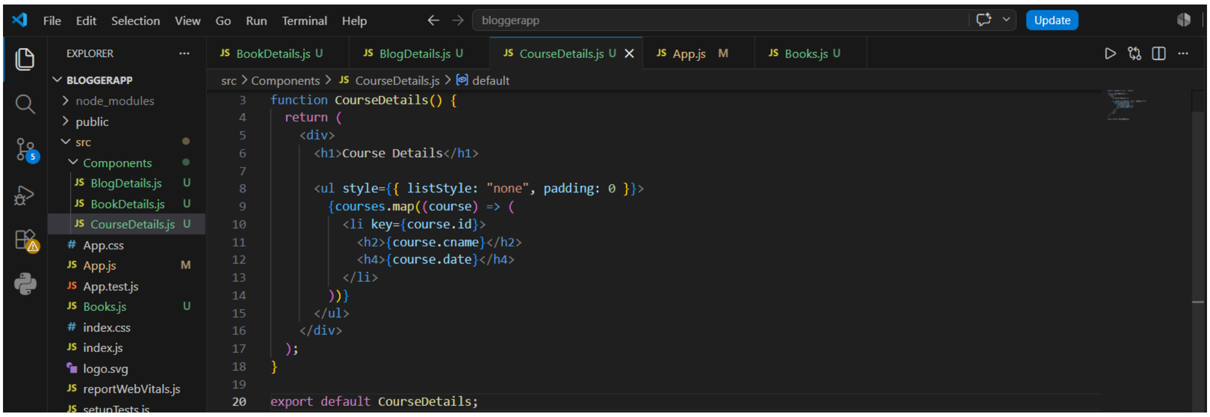



src > Components > BlogDetails.js > default
function BlogDetails() {

<h1>Blog Details</h1>
<ul style={{ listStyle: "none", padding: 0 }}>
{blogs.map((blog) => (
<li key={blog.id}>
<h3>{blog.title}</h3>
<h4>{blog.author}</h4>
{blog.description}

))}

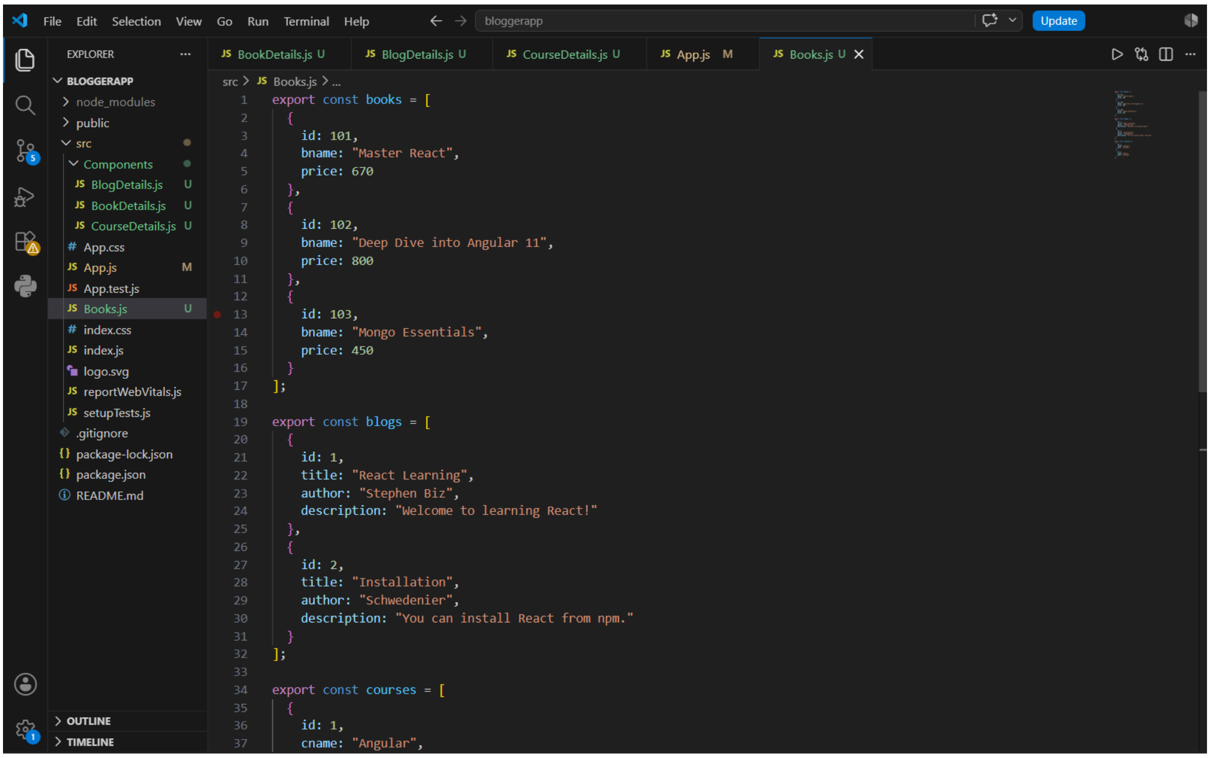
);
}
export default BlogDetails;


src > Components > CourseDetails.js > default
function CourseDetails() {
return (

<h1>Course Details</h1>
<ul style={{ listStyle: "none", padding: 0 }}>
{courses.map((course) => (
<li key={course.id}>
<h2>{course.cname}</h2>
<h4>{course.date}</h4>

))}

);
}
export default CourseDetails;


src > Books.js > ...
export const books = [
{
id: 101,
bname: "Master React",
price: 670
},
{
id: 102,
bname: "Deep Dive into Angular 11",
price: 800
},
{
id: 103,
bname: "Mongo Essentials",
price: 450
}
];
export const blogs = [
{
id: 1,
title: "React Learning",
author: "Stephen Biz",
description: "Welcome to learning React!"
},
{
id: 2,
title: "Installation",
author: "Schwedenier",
description: "You can install React from npm."
}
];
export const courses = [
{
id: 1,
cname: "Angular",

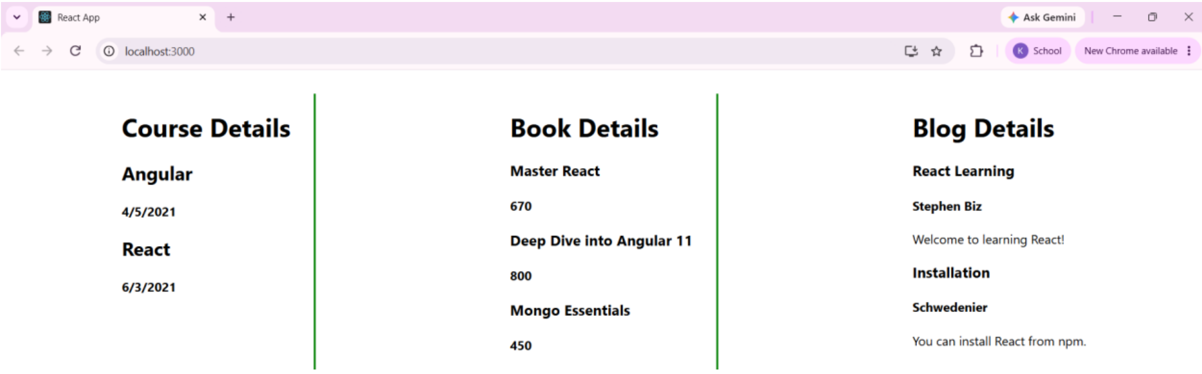
```
    date: "4/5/2021"
  },
  {
    id: 2,
    cname: "React",
    date: "6/3/2021"
  }
];
```

```
import "./App.css";
import BookDetails from "./Components/BookDetails";
import BlogDetails from "./Components/BlogDetails";
import CourseDetails from "./Components/CourseDetails";

function App() {
  return (
    <div
      style={{
        display: "flex",
        justifyContent: "space-around",
        padding: "30px"
      }}
    >
      <div
        style={{
          borderRight: "3px solid green",
          paddingRight: "30px"
        }}
      >
        <CourseDetails />
      </div>

      <div
        style={{
          borderRight: "3px solid green",
          paddingRight: "30px"
        }}
      >
        <BookDetails />
      </div>

      <div>
        <BlogDetails />
      </div>
    </div>
  );
```

```
}

export default App;
```

OUTPUT

```
Compiling...
Compiled successfully!

You can now view bloggerapp in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://192.168.21.1:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```


React App
Ask Gemini
localhost:3000
School
New Chrome available
Course Details
Angular
4/5/2021
React
6/3/2021
Book Details
Master React
670
Deep Dive into Angular 11
800
Mongo Essentials
450
Blog Details
React Learning
Stephen Biz
Welcome to learning React!
Installation
Schwedenier
You can install React from npm.